

1. General Description

The EMS3515 is a high-performance single-pole single-throw (SPST) depletion switch. Depletion technology enables the device to conduct signals in the absence of V_{DD} and to isolate signals in the presence of V_{DD} . The EMS3515 can pass or isolate negative signal fluctuation as low as -4.2 V and is 5.5 V tolerable. The EMS3515 's depletion gate control enables it to achieve exceptional THD+N performance while using very little power during signal conduction.

2. Features and Benefits

- Normally closed when $V_{DD} < 0.5$ V
- V_{SW} : - 4.2 V to 5.5 V
- R_{ON} : 0.85 Ω (typ.)
- $R_{FLAT(ON)}$: 8.7 m Ω (typ.)
- THD+N: - 102 dB (typ.)

3. Applications

- Smart phones, Tablets, laptops
- Bluetooth headsets, Mobile accessories

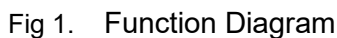
Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3515YFQ	E YW	WLCSP4	WLCSP package, 4 bumps, 0.93 mm × 0.93 mm; 0.4 mm pitch. 0.445 mm (Typ) height	3000

NOTE: YW = Date Code.

E Y W

- E** — Serial Number
- Y** — Date Code - Year
- W** — Date Code - Week

5. Function Diagram



6. Pinning Information

6.1. Pin map

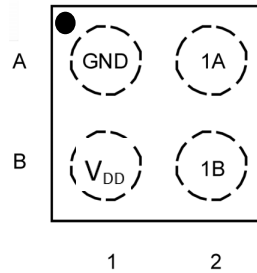


Fig 2. Top view pin configuration

6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
GND	A1	Ground
1A	A2	A-port of switch
V_{DD}	B1	Supply voltage (switch is conducted when V_{DD} is low)
1B	B2	B-port of switch

7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

V_{DD}	Switch state
L	ON (Conducting)
H	OFF (Isolating)

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions		Min	Max	Unit
V _{DD}	Supply voltage			- 0.5	6	V
V _{SW}	Switch voltage range	DC switch I/O voltage (switch conducting)		- 4.5	6	V
		DC switch I/O voltage (switch isolated)		- 4.5	6	V
I _{SW}	Maximum peak switch I/O current -Pulsed at 1ms duration, <10% duty cycle				500	mA
T _{stg}	Storage Temperature			- 65	150	°C
ESD	ESD Capability	HBM, JEDEC : JESD22-A114	I/O to GND	± 6		kV
			Others	± 7		kV

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _{DD}	Supply voltage	Isolating	2.0	5.5	V
V _{SW(OFF)}	Switch voltage	Isolating (V _{DD} =2.0V, R _L =32Ω, f<30kHz)		1.5	V _{rms}
		Isolating (V _{DD} =2.5V, R _L =32Ω, f<30kHz)		1.8	V _{rms}
		Isolating (V _{DD} =3.0V, R _L =32Ω, f<30kHz)		2.4	V _{rms}
		Isolating (V _{DD} =3.5V, R _L =32Ω, f<30kHz)		2.6	V _{rms}
		Isolating (V _{DD} =4.0V, R _L =32Ω, f<30kHz)		2.7	V _{rms}
		Isolating (V _{DD} =4.5V, R _L =32Ω, f<30kHz)		2.8	V _{rms}
		Isolating (V _{DD} =5.0V, R _L =32Ω, f<30kHz)		3.0	V _{rms}
		Isolating (requires 1 kΩ (typ.) in series with source)	4.7	5.5	V
V _{SW(ON)}	Switch voltage	Conducting (f<30kHz)		3.0	V _{rms}
		Isolating (requires 1 kΩ (typ.) in series with source)	4.7	5.5	V
T _A	Ambient Temperature		- 40	85	°C

10. Electrical Characteristics

Table 6. Static characteristics

Unless otherwise specified, typical values are at $V_{DD} = 0\text{ V}$, $T_A = +25\text{ }^{\circ}\text{C}$.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
I_{OFF}	Switch off leakage current	$V_{DD} = 3.3\text{ V to }4.5\text{ V}$, $1B = GND$	$1A = V_{DD}$	0.2		μA
			$1A = -1.5\text{ V to }1.5\text{ V}$	0.7		
		$V_{DD} = 2\text{ V to }3.3\text{ V}$, $1B = GND$	$1A = -1.5\text{ V to }1.5\text{ V}$	0.5		μA
R_{ON}	Switch on resistance	$I_{SW} = 50\text{ mA}$, $V_{SW} = -3\text{ V to }3\text{ V}$		0.85		Ω
$R_{FLAT(ON)}$	On resistance flatness	$I_{SW} = 50\text{ mA}$, $V_{SW} = -3\text{ V to }3\text{ V}$		8.7		$\text{m}\Omega$
C_{ON}	On capacitance	$V_{SW} = 0\text{ dBm}$, $f = 1\text{ MHz}$		17		pF
C_{OFF}	Off capacitance	$V_{SW} = 0\text{ dBm}$, $f = 1\text{ MHz}$		25		pF
AC Electronics Characteristics						
THD+N	Total harmonic distortion plus noise	$V_{SW} = 0.7\text{ V}_{RMS}$, Ground centered $R_L = 32\text{ }\Omega$, $f = 1\text{ kHz}$		-102		dB
OIRR	Off isolation rejection ratio	$V_{DD} = 2.0\text{ V}$ $V_{SW} < 1.7\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-100		dB
		$V_{DD} = 2.5\text{ V}$ $V_{SW} < 2.0\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-100		dB
		$V_{DD} = 3.0\text{ V}$ $V_{SW} < 2.4\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-100		dB
		$V_{DD} = 3.5\text{ V}$ $V_{SW} < 2.6\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-100		dB
		$V_{DD} = 4.0\text{ V}$ $V_{SW} < 2.7\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-108		dB
		$V_{DD} = 4.5\text{ V}$ $V_{SW} < 2.8\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-108		dB
		$V_{DD} = 5.0\text{ V}$ $V_{SW} < 3.0\text{ V}_{rms}$, $f = 20\text{ kHz}$	$R_L = 32\text{ }\Omega$	-115		dB
			$R_L = 1\text{ k}\Omega$	-108		dB

EMS3515

5 V SPST Depletion Switch with Negative Swing

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
BW	Bandwidth	$R_L = 50 \Omega$, $C_L = 0 \text{ pF}$		80		MHz
Supply currents						
I_{DD}	Quiescent current	$V_{DD} = 2.0 \text{ V}$		30		μA
		$V_{DD} = 2.5 \text{ V}$		37		μA
		$V_{DD} = 3.3 \text{ V}$		46		μA
		$V_{DD} = 4.2 \text{ V}$		62		μA
		$V_{DD} = 5.0 \text{ V}$		79		μA
I_{DIS}	Disable current	$V_{DD} \leq 0.2 \text{ V}$		0.1		μA
Control logic						
R_{PD}	V_{DD} pull-down resistance	$V_{DD} \leq 0.2 \text{ V}$		5.5		$\text{M}\Omega$
V_{DDH}	V_{DD} high voltage		2.0			V
V_{DDL}	V_{DD} low voltage				0.4	V
Switching characteristics						
t_{OFF}	Switch turn-off Time	$R_L = 1 \text{ k}\Omega$, $C_L = 10 \text{ pF}$, $V_{DD} = 0.0 \text{ V to } 2.0 \text{ V}$, $V_{SW} = 1.5 \text{ V}$;		170		μs
t_{ON}	Switch turn-on Time	$R_L = 1 \text{ k}\Omega$, $C_L = 10 \text{ pF}$, $V_{DD} = 2.0 \text{ V to } 0 \text{ V}$, $V_{SW} = 1.5 \text{ V}$;		1		μs
Oscillator frequency						
f_{OSC}	On-chip oscillator frequency	For reference only		3.3		MHz

11. Typical Characteristics

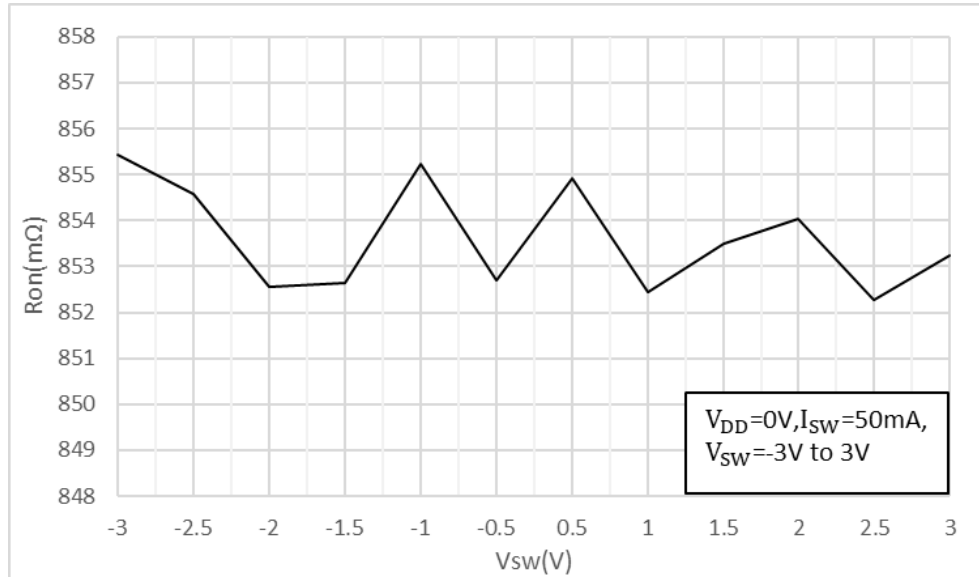


Fig 3. R_{ON} vs. V_{SW}

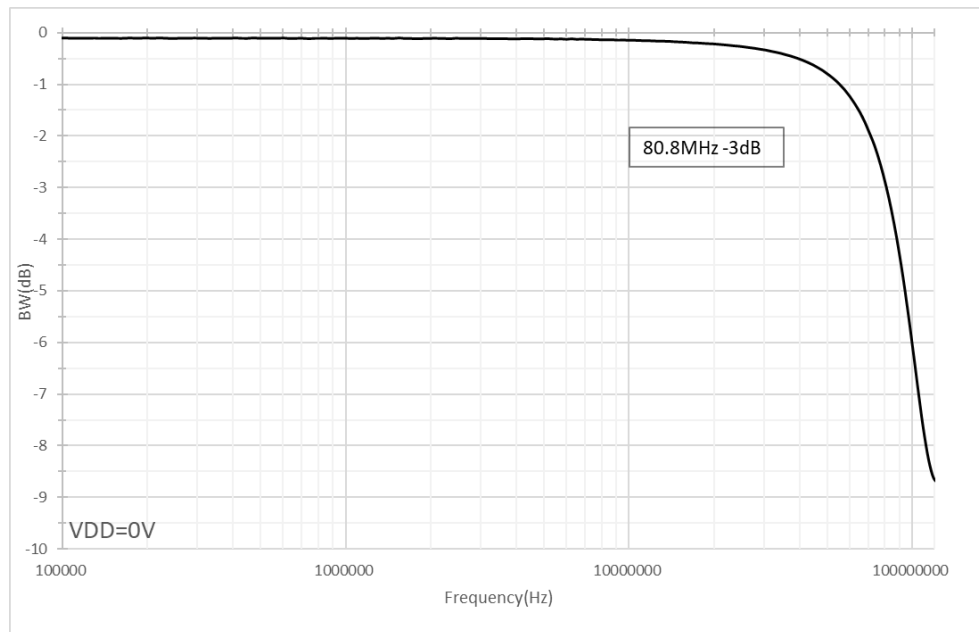


Fig 4. BW vs. Frequency

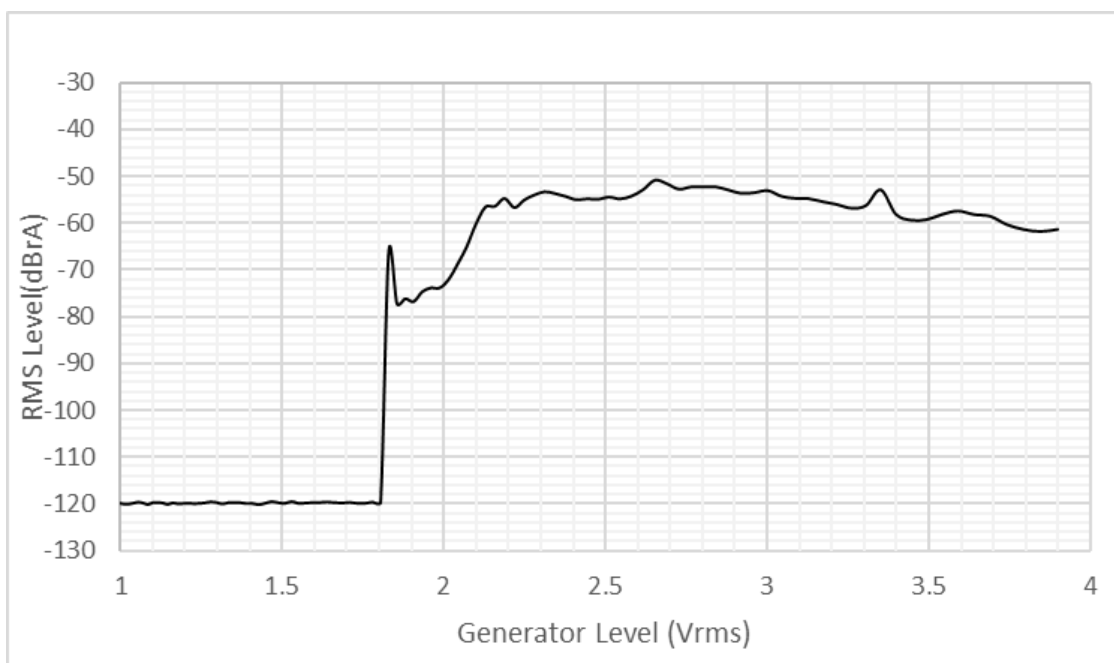


Fig 5. THD + N vs. Input with 33Ω Load

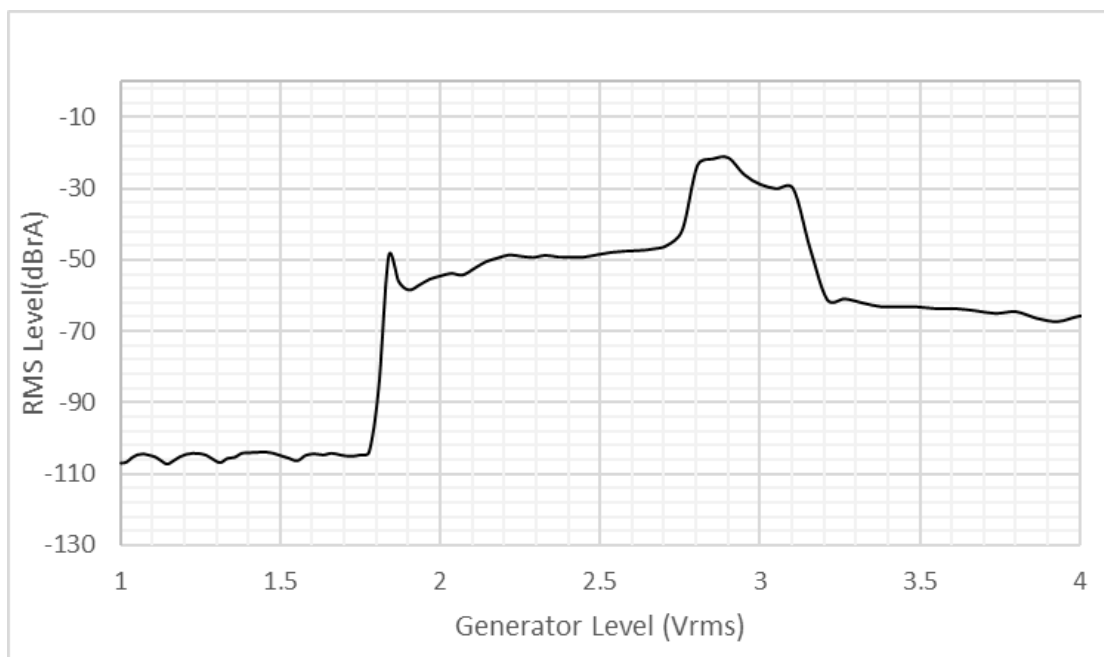


Fig 6. THD + N vs. Input with 1kΩ Load

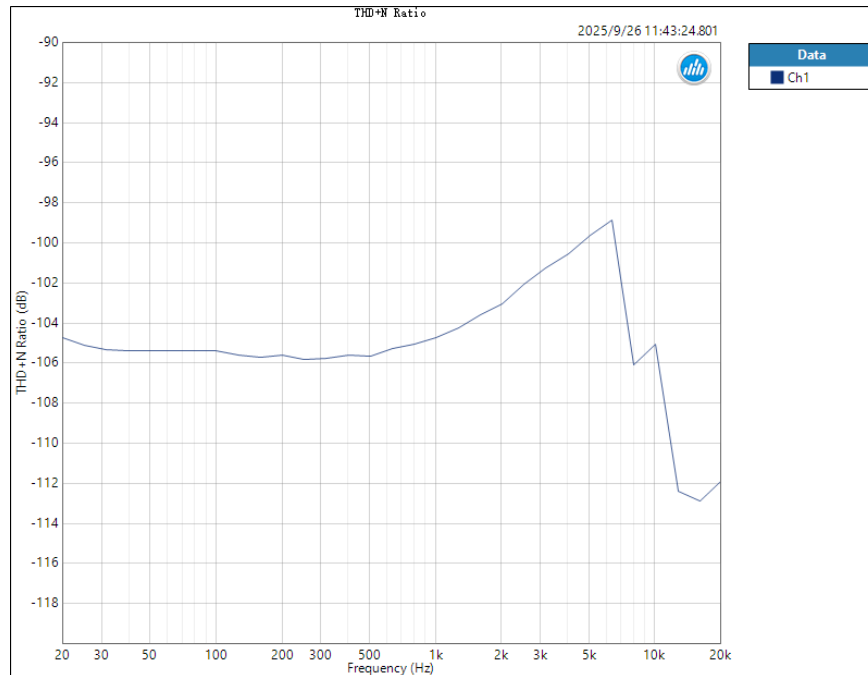


Fig 7. THD + N vs. Frequency with 33Ω Load

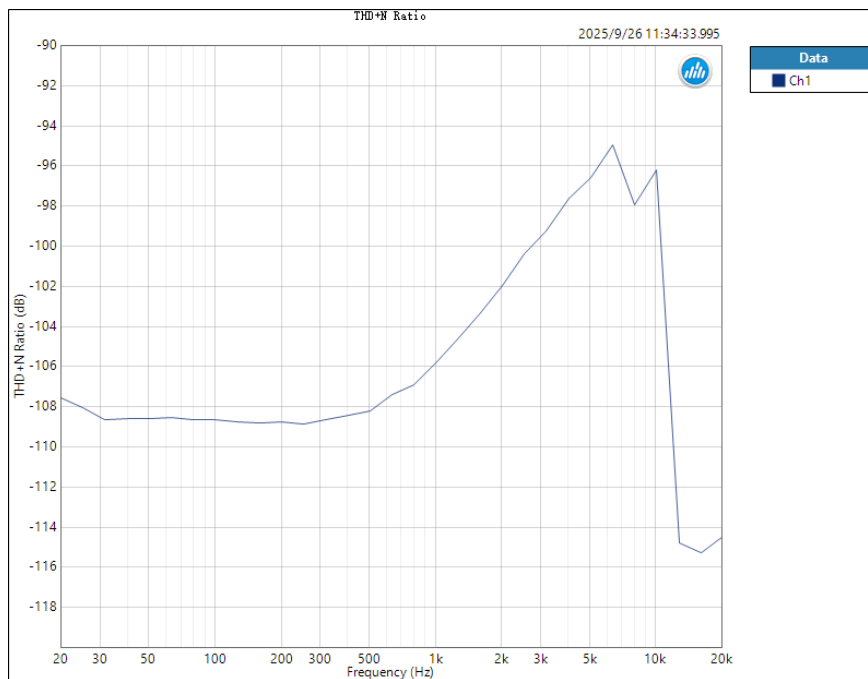


Fig 8. THD + N vs. Frequency with 1kΩ Load

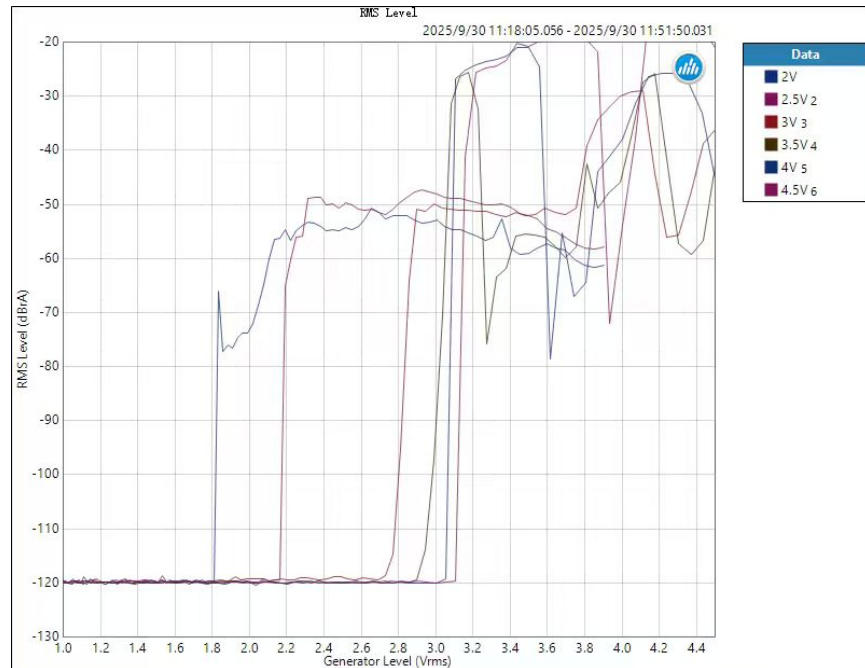


Fig 9. Off-isolation vs. Input with 32Ω Load

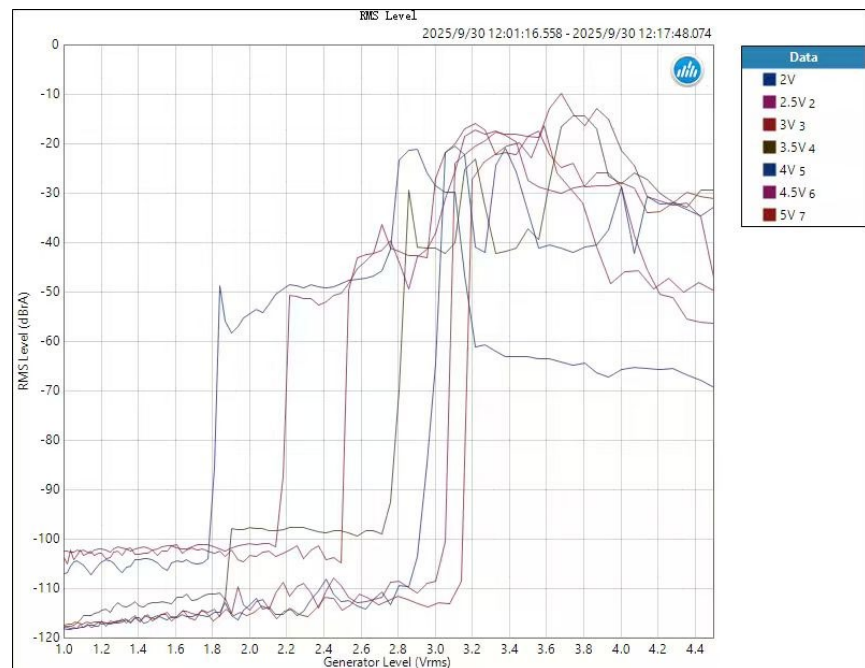
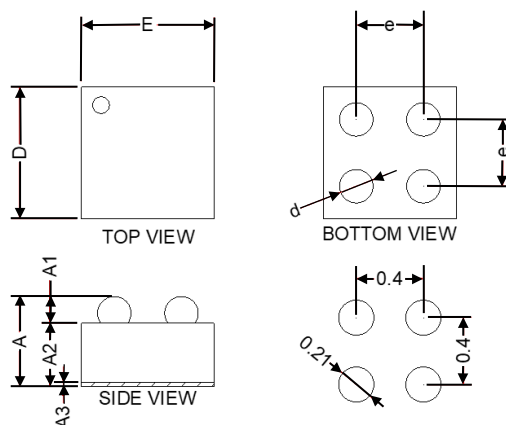


Fig 10. Off-isolation vs. Input with 1kΩ Load

12. Package Outline

WLCSP4



RECOMMENDED LAND PATTERN(unit:mm)

Symbol	Dimensions in Millimeters		
	Min.	Typ.	Max.
A	0.395	0.445	0.495
A1	0.144	0.16	0.176
A2	0.255	0.285	0.315
A3	0.02	0.025	0.03
D	0.9	0.93	0.96
E	0.9	0.93	0.96
e	0.35	0.4	0.45
d	0.19	0.21	0.23

13. Tape and Reel Information

13.1. Carrier tape dimensions

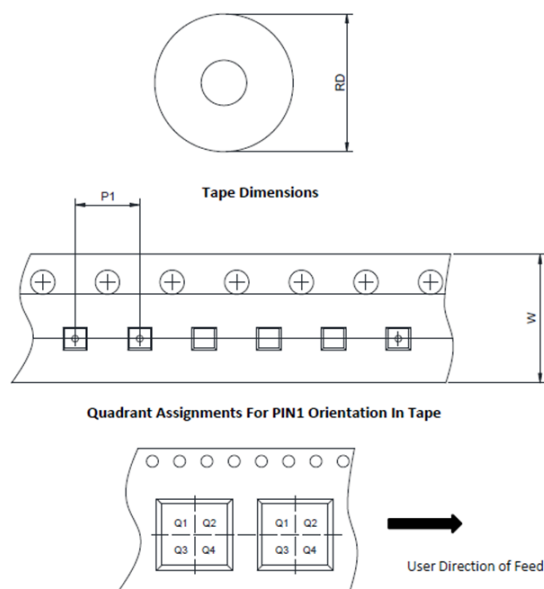


Table 7. Carrier tape dimensions

Package version	RD(mm)	P1(mm)	W(mm)	PIN 1
WLCSP4	180	4	8	Q1

13.2. Carrier tape dimensions

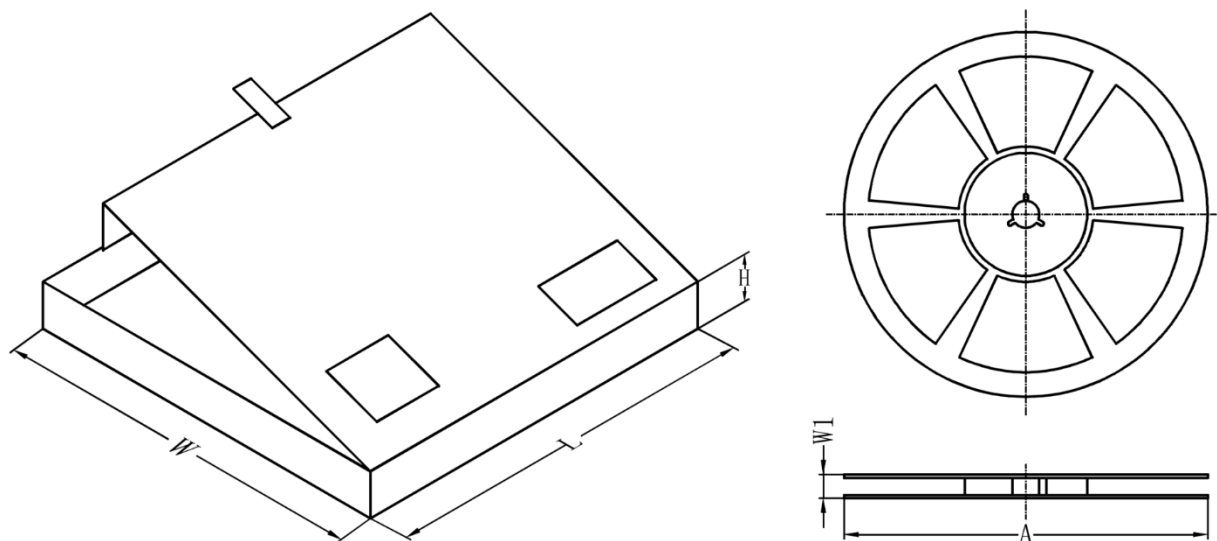


Table 8. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
WLCSP4	YFQ	180	12.2	3000	1	210x210x195

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

14. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3515 Rev. 1.2	Mar 04, 2026	Product datasheet		EMS3515 Rev. 1.1
Modifications:	- Table 4: V_{DD} and V_{SW} updated. - Table 5: V_{DD} and V_{SW} updated. - Section 4: Added topside marking information.			
EMS3515 Rev. 1.1	Nov 06, 2025	Product datasheet		EMS3515 Rev. 1.0
Modifications:	- Table 6 : Quiescent current I_{DD} and V_{DD} high voltage V_{DDH} updated. - Section 11: Added figures.			
EMS3515 Rev. 1.0	Oct 21, 2025	Product datasheet		